LEF MODEL · CHAPTER 2

Chapter 2: The Battle over the Length of the Day and the Angular-Momentum Balance

Based on Paper 02 — LEF Mechanism in Principle (sections A.4–A.8 and Part B)

01 · THE MYSTERY

The mystery: does the Moon run away from the finish line while running toward it?

In the previous chapter we discovered the finish line the material must reach to become the Moon: **the corotation radius**, which at the starting point (a 5.5-hour day) stood 15,818 km from the Earth's centre. But the moment the process begins and the pear lengthens upward, an intriguing physical trap appears.

The law of conservation of angular momentum says that the total amount of rotation of an isolated system cannot be created out of nothing. When the Earth lifts a large mass far from its axis and forces it to keep turning with it at the same rate, the rising mass "steals" angular momentum from the Earth's rotation. As a result the Earth's rotation slows, the day lengthens — **and the corotation line itself retreats and moves outward!**

Here three decisive questions are born that settle the fate of the whole model:

01

First, **does the lengthening of the day drive the corotation line too far away**, or does an internal process take place at the same time that speeds up the rotation and

balances it?

02

Second, **is the total angular-momentum budget of the early Earth enough** to pay the "tax" of lifting the Moon, and still leave us with exactly the angular momentum measured in the Earth–Moon system today?

03

And third, **what prevents the young Moon from being torn to pieces** by the Earth's enormous tidal forces at the moment of separation?

02 · AN IMAGE

An image: the figure skater who stretches out one arm — and folds in the other

We all know the classic sight of a figure skater spinning on the spot: when she spreads her arms to the sides, her spin slows at once; and when she pulls her arms in to her body, she spins at a dizzying speed. The reason is that mass far from the axis of rotation increases the moment of inertia, and mass close to the axis decreases it.

Now imagine a special acrobatic routine in which the skater performs **two opposite actions at the same time**: with one hand she reaches out a long, heavy pole (the "pear" that lengthens toward the corotation line and tries to slow her), and at the same time she pulls inward, toward the centre of her body, a heavy iron weight she had earlier held far from the axis (the "core" that sinks toward the Earth's centre and speeds her up).

Two opposing forces

Because these two opposing forces act together, the skater hardly loses any of her spin rate! She slows only slightly, and stays steady at exactly the rate needed to complete the routine.

Science and model: why a "uniform interior" is a threshold condition

Many models of the Earth's evolution tend to assume that its heavy iron core formed and sank to the centre at top speed, even before the Moon was formed. But Paper 02 shows that such an assumption blocks the angular-momentum budget: when all the iron is concentrated at the centre, the Earth's moment of inertia is small (the inertia factor falls from $k = 0.4$ for a uniform sphere to $k \approx 0.33$ for today's differentiated Earth), and with the same 5.5-hour day it cannot store enough angular momentum.

In the LEF model, by contrast, the organisation of the core is not a one-time event that ended beforehand, but **a single continuous process**:

> **At the start of the mass transfer:** the young Earth was hot, swollen (a thermal expansion of 120 km, within the allowed range of 98–520 km) and still almost uniform in its mass distribution ($k \gtrsim 0.397$). In this state it stored an initial angular-momentum budget of $L_0 \simeq 3.466 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}$.

> **During the rise of the pear:** the heavy iron gradually sank inward to the centre (in the two-layer model: a radius ratio $q = 0.55$ and a density ratio $D = 2$), producing exactly the speed-up of the rotation needed to balance the lengthening of the pear!

The key to the price in angular momentum

In Jacobi's classical fission route (in which the Earth tries to spin up its own surface to the breaking point, a day of 2.73 hours), the system needs $9.42 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}$ — that is, **2.74 times** the angular momentum of today's Earth–Moon system (and at the original reference point 4.08 times). In the LEF model, where the material is carried to the corotation line, the initial budget of the parent body stands at **1.007 times** today's angular momentum — a tiny surplus of only 0.7% (and with vector addition at the angles tested: 1.5%–2.5%), which matches exactly the later loss to the solar tide!

4.08	2.74	1.007
Classical fission · original reference point	Classical fission · Jacobi bifurcation point	LEF model · release at the corotation line

04 · THE MECHANISM IN WORDS

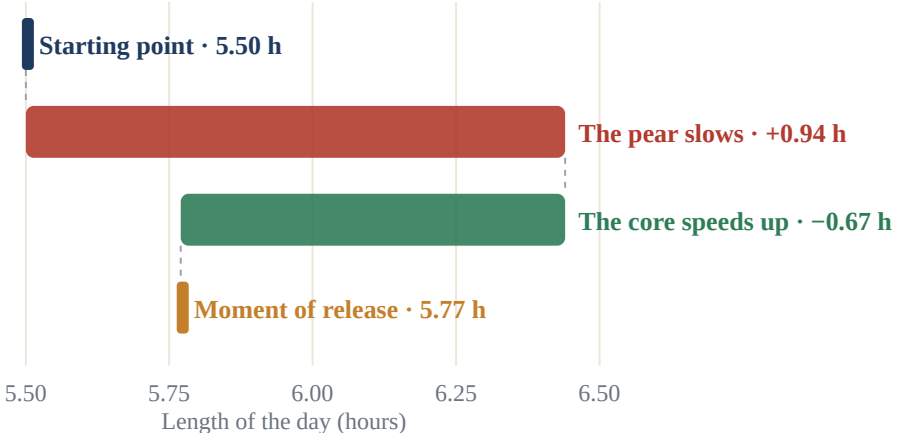
The mechanism in words: the battle over the day, the Roche limit and the watershed

Let us follow, step by step, the outcome of the fascinating battle that takes place inside the Earth while the pear rises:

1. **The pear tries to slow the Earth (+0.94 hours):** transferring mass outward in corotation gives the rising material about 15.6% of the angular-momentum budget in the basic calculation (and about 16.5% in the coupled solution). This angular momentum is taken from the Earth's rotation. Had the pear acted alone, without the organisation of the core ($D = 1$), the day would have lengthened from 5.50 to **6.44 hours** and the corotation line would have run out to **17,569 km**.

2. **The sinking core speeds up the Earth (−0.67 hours):** at the same time, the heavy iron sinks to the Earth's centre (in the two-layer scenario $q = 0.55$, $D = 2$) and shortens the day by **0.67 hours**.

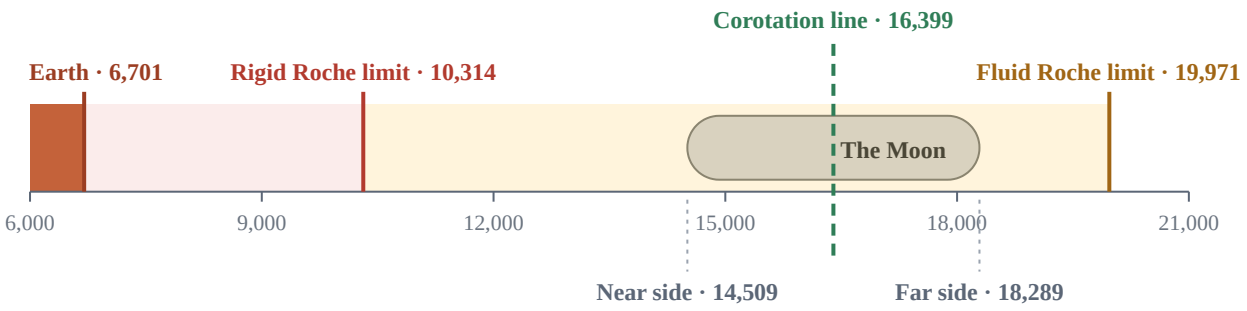
3 **The point of balance at the moment of release (5.77 hours and 16,399 km):** when the two processes act together, the speed-up from the core offsets most of the slowing by the pear. The day at the moment of separation stands at only **5.77 hours**, and the corotation line settles at a distance of **16,331 km** (in the test-particle approximation) or **16,399 km** (≃ 16,400 km) when the Moon's own mass is included in a two-body problem.



The battle over the length of the day: the pear alone would have lengthened it to 6.44 hours; the sinking core gives back 0.67 hours, and the day at the moment of release is 5.77 hours.

> How does the Moon survive the Roche limit and begin to move away at once?

At the moment of release, the Moon is a round, rigid body (radius $R_m = 1{,}890$ km at a density of 2,600 kg/m³): its near side stands 14,509 km from the Earth's centre (about 7,800 km above the surface of the flattened Earth, whose equatorial radius is 6,701 km), and its far side at 18,289 km. This position lies inside the **fluid Roche limit** (19,971 km) — the distance within which the Earth's tidal forces would tear a soft fluid sphere to pieces — but far outside the **rigid Roche limit** (10,314 km). Because the Moon had already completed its rounding and developed a rigid outer shell ($\mu \geq 10$ GPa) while still attached to the pear, its axis ratio stays at most 1.04 (almost a perfect sphere) and it survives intact!



Distance from the Earth's centre (km) · the Moon at release, radius 1,890 km

The Moon at the moment of release on the distance axis: outside the rigid Roche limit and inside the fluid Roche limit, with the corotation line passing through it.

And finally, why does it move away at once instead of falling back? Because the corotation line is a **tidal watershed**: a body sitting exactly on the line is in neutral equilibrium; a body inside the line is dragged inward by the tide; while a body released **slightly beyond the line** — with a margin of $\Delta r \simeq 2d_0 \approx 210$ km beyond the corotation line, when about 58% of the Moon's volume has already crossed it — is pushed forward and outward by the tidal bulge of the Earth, which spins faster than the Moon, and begins to move away from the very first orbit!

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The system's angular-momentum budget (L_0) is conserved between the initial state and the moment of release, and is divided between the Earth's rotation and the Moon's orbit:

Conservation of angular momentum

$$L_0 = I_0\omega_0 \simeq 3.466 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}} = I_\oplus\Omega$$

The corotation line at the moment of release (5.77 h)

$$r = R_{\mathrm{sync}} = \left(\frac{G(M_\oplus + m)}{\Omega^2}\right)^{1/3} \simeq 16{,}399\ \mathrm{k}$$

At the moment of separation, the Moon's orbit holds $5.9 \times 10^{33}\ \mathrm{kg\,m^2\,s^{-1}}$ (17% of the budget, with the reduced mass μ_{red}; or 6.0×10^{33} computing with the full mass

around the Earth's centre), and the Earth's rotation holds $2.88 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}$. The moment-of-inertia factor derived for the Earth at this moment is $k = 0.355$ — a striking match to the value $k = 0.360$ of a two-layer structure ($q = 0.55$, $D = 2$):



The division of the angular-momentum budget at the moment of release.

The two-layer inertia factor

$$k = \frac{2}{5}\,\frac{(1-q^5)+Dq^5}{(1-q^3)+Dq^3} = 0.360$$

The rigid Roche limit

$$d_{\mathrm{rigid}} = 1.26\,R_\oplus \left(\frac{\rho_\oplus}{\rho}\right)^{1/3} \simeq 10{,}314\ \mathrm{km}$$

06 · THE SIGNATURE

The signature in reality: Phobos and Deimos around Mars

The idea that the corotation line acts as a tidal "watershed" is not a theoretical hypothesis — it is on live display in our Solar System, around the planet **Mars**. Mars has two small moons orbiting it on either side of the Martian corotation radius (20,428 km):

> **Deimos (the outer moon, 23,463 km):** lies *outside* the corotation line of Mars (it orbits more slowly than Mars rotates). So the tide of Mars pushes it outward, and it gradually moves away from Mars — just like our Moon, which was released about 210 km beyond the corotation line.

> **Phobos (the inner moon, about 9,376 km):** lies *inside* the corotation line (it runs ahead of the rotation of Mars). So the tide of Mars brakes it and pulls it inward, and it is gradually falling toward a crash on the surface of Mars in tens of millions of years.



The tidal watershed around Mars: Phobos inside the line and pulled inward, Deimos outside it and moving away.

The lab box: the balance of the day, the angular momentum and the Roche limits (Paper 02, Part B)

Stage / parameter	Length of the Earth's day	Radius of the corotation line	Meaning and division of the budget
1. The starting point	**5.50 hours**	15,818 km	Initial budget $L_0 = 3.466 \times 10^{34}$ kg m^2 s^{-1} ($k \gtrsim 0.397$)
2. The pear lengthens alone ($D = 1$)	**6.44 hours (+0.94 h)**	17,569 km	Slowing due to the transfer of 15.6%–16.5% of the angular momentum to the rising material
3. The core sinks ($D = 2$)	**−0.67 hours**	Pulled inward	A speed-up ("the skater pulling in her arms") that offsets most of the slowing
4. The combined release state	**5.77 hours**	**16,399 km (16,331 test particle)**	**The budget closes at 1.007 × today (17% in the orbit, 83% in the Earth's rotation)**
5. The release margin and Roche	**5.77 hours**	Release at +210 km beyond the line	A rigid Moon ($\mu \geq 10$ GPa, axis ratio ≤ 1.04) outside the rigid Roche limit (10,314 km)

07 · NEXT

The door ahead

We have shown that the mechanics of nature allows it: the angular-momentum budget closes to within 1.007, the battle between the pear and the sinking core steadies the day at 5.77 hours, and the Moon is released rigid and round beyond the corotation line and begins to move away at once.

But Paper 02 proved only that the route is possible *in principle*. Now the question of execution arises: **what enormous engine pushed billions of tonnes of rock to a height of thousands of kilometres against gravity?** In the next chapter — **Chapter 3: The Engine Within — the Source Pressure of 150 GPa and the Foamed Magma** (from Paper 03 — *LEF Mechanism in Practice*) — we will go down into the depths of the early Earth and meet the pressure that lifted the Moon.

Chapter 3

The Engine Within — the Source Pressure of 150 GPa and the Foamed Magma →

← Chapter 1 · Prologue

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